

Probability Review 1

Discrete Random Variables

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Roadmap

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2. Sample Space and Events
3. Set Operations on Events
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1. Motivation: Probability vs. Statistics

1.1 Why Probability?

This course is about **statistics**: using observed data to learn something about an unobserved process that generated the data.

But we cannot do statistics without first understanding **probability**, the mathematical language for describing random processes.

The two halves of the same coin

- **Probability:** given a model of randomness, compute the chance of an event.
- **Statistics:** given observed events (data), recover information about the underlying model.

This first lecture is a probability primer. We will use these tools every week of the semester.

2. Sample Space and Events

2.1 Sample Space

Definition

The **sample space** S of a random experiment is the set of all possible outcomes.

Some examples (we'll keep coming back to these):

- Coin toss: $S = \{\text{Heads}, \text{Tails}\}$.
- Letter grade: $S = \{A, B, C, D, E\}$.
- Roll one die: $S = \{1, 2, 3, 4, 5, 6\}$.
- Toss two coins: $S = \{HH, HT, TH, TT\}$.

Curly braces $\{\cdot\}$ denote a set; the symbols inside are the possible outcomes.

2.2 Events

Definition

An **event** A is any subset of the sample space, written $A \subseteq S$. The **probability** of A , written $\mathbb{P}(A)$, is a number in $[0, 1]$ satisfying

$$\mathbb{P}(S) = 1, \quad 0 \leq \mathbb{P}(A) \leq 1.$$

Notation reminder. The symbol $\subseteq$ means “is a subset of”: $A \subseteq S$ says every outcome listed in A is also in S . We write $\mathbb{P}(\cdot)$ for “probability of”; it always returns a number between 0 and 1.

Examples:

- Roll one die. Let $A = \{2, 4, 6\}$ (“rolled an even number”). Then $\mathbb{P}(A) = 3/6 = 1/2$.
- Toss two coins. Let $A =$ “at least one heads” $= \{HH, HT, TH\}$. Then $\mathbb{P}(A) = 3/4$.

2.3 Complement

Definition

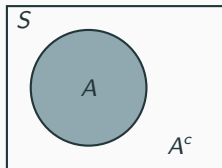
The **complement** of an event A , written A^c , is the set of outcomes *not* in A . Equivalently, $A^c = S \setminus A$ (“ S with A removed”).

Result

$$\mathbb{P}(A) + \mathbb{P}(A^c) = 1.$$

Why? An outcome is either in A or not in A , never both. So the chances of the two cases must add to 1.

Example: roll a die, $A = \{2, 4, 6\}$. Then $A^c = \{1, 3, 5\}$, and $\mathbb{P}(A^c) = 1 - \frac{1}{2} = \frac{1}{2}$.



2.4 Example: Self-Checkout

Setup. 40% of customers at a grocery store use the self-checkout lane. What is the probability that a randomly selected customer does *not* use self-checkout?

Define the event. Let A = “customer uses self-checkout.” We are told $\mathbb{P}(A) = 0.40$.

Compute. The complement A^c = “customer does not use self-checkout.” Apply $\mathbb{P}(A) + \mathbb{P}(A^c) = 1$:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A) = 1 - 0.40 = 0.60.$$

Interpretation. 60% of customers don't use self-checkout.

3. Set Operations on Events

3.1 Intersection and Union

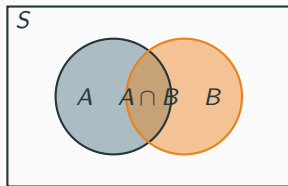
Definitions

- **Intersection** $A \cap B$: outcomes in *both* A and B .
- **Union** $A \cup B$: outcomes in A , in B , or in both.

Notation reminder. Read $\cap$ as “and” (intersection) and $\cup$ as “or, possibly both” (union).

Quick check. Roll a die; $A = \{2, 4, 6\}$, $B = \{5, 6\}$:

- $A \cap B = \{6\}$ (only 6 is in both).
- $A \cup B = \{2, 4, 5, 6\}$ (everything in either).



Shaded overlap: $A \cap B$.

Whole shaded region: $A \cup B$.

3.2 Inclusion–Exclusion Formula

Key Result

For any two events A and B ,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Why subtract $\mathbb{P}(A \cap B)$? Because $\mathbb{P}(A) + \mathbb{P}(B)$ counts the overlap (the part inside both circles) twice, once in each term. Subtracting $\mathbb{P}(A \cap B)$ corrects the double count.

We apply this formula in the worked card example below.

3.3 Mutually Exclusive Events

Definition

Events A and B are **mutually exclusive** (or **disjoint**) if $A \cap B = \emptyset$, i.e., they cannot both occur.

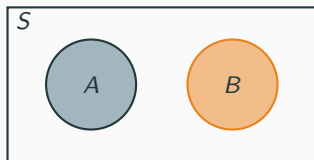
Notation reminder. The symbol $\emptyset$ stands for the empty set — the set with no outcomes in it.

In the disjoint case the formula simplifies:

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

(No overlap means nothing to subtract.)

Examples: “Pass / Fail the class.” “Roll a 1 / Roll a 2.”



3.4 Example: Drawing One Card

Setup. Draw one card from a standard 52-card deck. Define

- K = “card is a King”,
- R = “card is red”,
- C = “card is a club”.

Step 1: count.

$$\mathbb{P}(K) = \frac{4}{52}, \quad \mathbb{P}(R) = \frac{26}{52}.$$

Two of the four Kings (hearts, diamonds) are red, so $\mathbb{P}(K \cap R) = \frac{2}{52}$.

Step 2: union by inclusion-exclusion.

$$\mathbb{P}(K \cup R) = \mathbb{P}(K) + \mathbb{P}(R) - \mathbb{P}(K \cap R) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52}.$$

3.4 Example: Drawing One Card 2

Step 3: a disjoint pair. A red card is never a club, so R and C are mutually exclusive: $R \cap C = \emptyset$ and $\mathbb{P}(R \cap C) = 0$.

Interpretation. About 54% of cards are either a King or red (or both). Red and club are mutually exclusive — there are no overlap cards.

4. Conditional Probability and Independence

4.1 Motivation: Information Changes Probability

Setup. You catch the bus to class with probability 0.9 on a normal day. But if you wake up late, your chance drops to 0.4.

Today you happen to know you woke up late. What probability should you now assign to making your bus?

Two probabilities to keep separate:

- The unconditional chance of making the bus: 0.9.
- The chance of making the bus *given* that you woke up late: 0.4.

Knowing extra information (you woke up late) shifts the relevant probability. We need a formal way to talk about this.

4.2 Conditional Probability

Definition

The **conditional probability** of A given B (with $\mathbb{P}(B) > 0$) is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Notation reminder. The vertical bar $|$ is read “given”: $\mathbb{P}(A \mid B)$ is the probability of A under the assumption that B has already happened.

Intuition. Once we know B has occurred, the relevant sample space shrinks from S to B . Within that smaller world, the fraction occupied by A is the overlap $A \cap B$ divided by B .

4.3 Multiplication Rule

Rearranging the conditional definition gives the **multiplication rule**:

$$\mathbb{P}(A \cap B) = \mathbb{P}(B) \mathbb{P}(A \mid B).$$

By symmetry, we can also write

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B \mid A).$$

This is how we move between joint probabilities and conditional probabilities.

4.4 Example: Scholarship Applications

Setup. A school accepts 10% of applicants. Of those accepted, 2% receive a scholarship. What is the probability that a random applicant gets accepted *and* receives a scholarship?

Define events. Let A = “accepted” and S = “receives scholarship.” We are told

$$\mathbb{P}(A) = 0.10, \quad \mathbb{P}(S \mid A) = 0.02.$$

Note: $\mathbb{P}(S \mid A) = 0.02$ is a *conditional* probability — it’s the chance of a scholarship *given* you’ve already been accepted.

Apply the multiplication rule.

$$\mathbb{P}(S \cap A) = \mathbb{P}(A) \mathbb{P}(S \mid A) = 0.10 \times 0.02 = 0.002.$$

Interpretation. Only 0.2% of applicants — 2 out of every 1,000 — both get in and receive a scholarship.

4.5 Independence

Definition

Two events A and B are **independent** if knowing one does not change the probability of the other:

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Equivalent characterization

A and B are independent iff

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Why equivalent? Plug the multiplication rule $\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B)$ and the assumption $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ together: $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Where this shows up later. The *i.i.d.* assumption (independent and identically distributed) is the foundation of nearly every estimator we'll build: OLS (Lectures 6–9), classification (Lecture 12), resampling (Lecture 13), and trees/forests (Lecture 14).

4.6 Example: Are Two Failures Independent?

Setup. Suppose we observe device crash rates:

$$\mathbb{P}(\text{smartphone crash}) = 0.02, \quad \mathbb{P}(\text{laptop crash}) = 0.06,$$

and the chance both crash on the same day is 0.0012.

Question. Are the two crashes independent events?

Set up. Let L = “laptop crash” and S = “smartphone crash.” We are given $\mathbb{P}(L) = 0.06$, $\mathbb{P}(S) = 0.02$, $\mathbb{P}(S \cap L) = 0.0012$.

Test. Compute $\mathbb{P}(S \mid L)$ and compare to $\mathbb{P}(S)$:

$$\mathbb{P}(S \mid L) = \frac{\mathbb{P}(S \cap L)}{\mathbb{P}(L)} = \frac{0.0012}{0.06} = 0.02 = \mathbb{P}(S).$$

Conclusion. Since $\mathbb{P}(S \mid L) = \mathbb{P}(S)$, knowing the laptop crashed tells us nothing about the smartphone — the events are independent.

4.7 Independent vs. Disjoint — Don't Confuse Them!

These two ideas sound similar but are very different.

- **Disjoint:** $A \cap B = \emptyset$, so $\mathbb{P}(A \cap B) = 0$. The two events cannot both happen.
- **Independent:** $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Knowing one tells you nothing about the other.

In fact, if A and B are disjoint and both have positive probability, they are *not* independent: knowing A occurred tells you that B did *not* occur, which is highly informative. Disjoint events are maximally dependent on each other.

5. Total Probability and Bayes' Theorem

5.1 Partition of the Sample Space

Definition

A collection of events $A_1, A_2, \dots, A_J$ forms a **partition** of S if:

- they are pairwise disjoint: $A_i \cap A_j = \emptyset$ for $i \neq j$, and
- they cover S : $\bigcup_{j=1}^J A_j = S$.

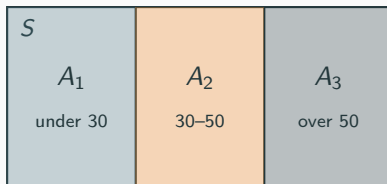
Notation reminder. The big union $\bigcup_{j=1}^J A_j$ means $A_1 \cup A_2 \cup \dots \cup A_J$ — chain together every set in the list.

5.2 Example of a Partition

Setup. Categorize every customer entering a store by age bracket:

$$A_1 = \text{under 30}, \quad A_2 = 30\text{--}50, \quad A_3 = \text{over 50}.$$

Every customer is in exactly one bracket: the brackets don't overlap (disjoint) and together they cover all customers (union equals S). So $\{A_1, A_2, A_3\}$ is a partition.



Simplest possible partition: any event A together with A^c always forms a partition of S .

5.3 Law of Total Probability

Key Result

Let $A_1, \dots, A_J$ partition S . For any event B ,

$$\mathbb{P}(B) = \sum_{j=1}^J \mathbb{P}(B \cap A_j) = \sum_{j=1}^J \mathbb{P}(A_j) \mathbb{P}(B \mid A_j).$$

Notation reminder. The sigma symbol $\sum_{j=1}^J$ is shorthand for “add up the term for $j = 1$, then $j = 2$, all the way to $j = J$.” So $\sum_{j=1}^3 a_j = a_1 + a_2 + a_3$.

In words: the probability of B is a weighted average of the conditional probabilities $\mathbb{P}(B \mid A_j)$, with weights $\mathbb{P}(A_j)$.

5.3 Law of Total Probability 2

Special case ($J = 2$, partition A and A^c):

$$\mathbb{P}(B) = \mathbb{P}(A)\mathbb{P}(B \mid A) + \mathbb{P}(A^c)\mathbb{P}(B \mid A^c).$$

5.4 Example: Job Interview

Setup. If your interview goes well, you have a 70% chance of getting the job. If it goes badly, you have a 10% chance. You think the interview is equally likely to go well or badly. What's your overall probability of getting the job?

Build a partition. Let I_1 = “interview goes well”, I_2 = “interview goes badly.” Together I_1, I_2 partition the sample space (every interview either does or doesn't go well). Let J = “get the job.” We are given:

$$\mathbb{P}(I_1) = 0.5, \quad \mathbb{P}(I_2) = 0.5, \quad \mathbb{P}(J \mid I_1) = 0.7, \quad \mathbb{P}(J \mid I_2) = 0.1.$$

Apply the Law of Total Probability.

$$\begin{aligned}\mathbb{P}(J) &= \mathbb{P}(I_1)\mathbb{P}(J \mid I_1) + \mathbb{P}(I_2)\mathbb{P}(J \mid I_2) \\ &= 0.5 \times 0.7 + 0.5 \times 0.1 \\ &= 0.35 + 0.05 = 0.40.\end{aligned}$$

Interpretation. Overall, you have a 40% chance of getting the job.

5.5 Motivating Example: A Medical Test

Setup. A disease affects 1% of the population. A test for it is 99% accurate when you are sick (correctly returns positive) and 95% accurate when you are healthy (correctly returns negative).

You test positive. What is the probability you actually have the disease?

Most people guess something close to 99%. We will see the answer is about 17%. To get there we need a tool that lets us flip a conditional — to go from $\mathbb{P}(\text{positive} \mid \text{sick})$ (which we know) to $\mathbb{P}(\text{sick} \mid \text{positive})$ (what we want).

That tool is Bayes' theorem.

5.6 Bayes' Theorem

Bayes' Theorem

For any events A, B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A) \mathbb{P}(A)}{\mathbb{P}(B)}.$$

Derivation. The multiplication rule lets us write $\mathbb{P}(A \cap B)$ two ways:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B \mid A) = \mathbb{P}(B) \mathbb{P}(A \mid B).$$

Solve the second equality for $\mathbb{P}(A \mid B)$ and you get Bayes.

In practice the denominator $\mathbb{P}(B)$ is computed via the Law of Total Probability over a partition $A_1, \dots, A_J$:

$$\mathbb{P}(A_k \mid B) = \frac{\mathbb{P}(B \mid A_k) \mathbb{P}(A_k)}{\sum_{j=1}^J \mathbb{P}(B \mid A_j) \mathbb{P}(A_j)}.$$

This is the workhorse form for updating beliefs when several mutually exclusive causes could have produced an observation (medical testing,

5.7 Resolving the Medical Test Example

Define events. Let D = “has the disease”, T = “test is positive.” We are given:

$$\mathbb{P}(D) = 0.01, \quad \mathbb{P}(T \mid D) = 0.99, \quad \mathbb{P}(T \mid D^c) = 1 - 0.95 = 0.05.$$

We want $\mathbb{P}(D \mid T)$.

Step 1: total probability of testing positive. Use the partition $\{D, D^c\}$:

$$\begin{aligned}\mathbb{P}(T) &= \mathbb{P}(T \mid D)\mathbb{P}(D) + \mathbb{P}(T \mid D^c)\mathbb{P}(D^c) \\ &= 0.99 \times 0.01 + 0.05 \times 0.99 \\ &= 0.0099 + 0.0495 = 0.0594.\end{aligned}$$

About 5.9% of all tests come back positive.

5.7 Resolving the Medical Test Example 2

Step 2: apply Bayes' theorem.

$$\begin{aligned}\mathbb{P}(D \mid T) &= \frac{\mathbb{P}(T \mid D) \mathbb{P}(D)}{\mathbb{P}(T)} \\ &= \frac{0.99 \times 0.01}{0.0594} \\ &\approx 0.167.\end{aligned}$$

Interpretation. Even with a 99%-accurate test, a positive result only gives about a 17% chance you actually have the disease.

Why so low? Because the disease is rare. Out of 10,000 people, about 100 are sick (and 99 test positive); about 9,900 are healthy (but $5\% \times 9,900 = 495$ false positives). So most positive tests come from healthy people.

6. Random Variables

6.1 What is a Random Variable?

Definition

A **random variable** (RV) is a function

$$X : S \longrightarrow A \subseteq \mathbb{R}$$

that assigns a real number $X(s)$ to every outcome s in the sample space S . For a discrete RV, the set A is a finite or countable subset of $\mathbb{R}$ (e.g., $\{0, 1, 2, \dots\}$).

Notation reminder. A **function** $f : A \rightarrow B$ is a rule that takes each input from A and returns exactly one output in B . So $X : S \rightarrow A$ says: “feed me an outcome s from the sample space, and I give you back a number $X(s)$.” Many objects in this course are functions (PMFs, CDFs, densities, the regression function $f(x)$, loss functions). Get comfortable with the idea early.

Random variables are denoted by capital letters (X, Y, Z); the realized value is denoted by the corresponding lowercase letter (x, y, z).

6.2 Probability Mass Function (PMF)

Let X be a discrete random variable.

Definition

The **probability mass function (PMF)** of X is

$$p_X(x) = \mathbb{P}(X = x).$$

It gives the probability that X takes the specific value x .

Notation reminder. The subscript X in p_X tells us “this PMF belongs to the random variable X .” When there’s only one RV around we sometimes drop the subscript and write $p(x)$.

Properties

- $0 \leq p_X(x) \leq 1$ for every value x .
- $\sum_x p_X(x) = 1$, where the sum is over all values X can take.

6.3 Cumulative Distribution Function (CDF)

Definition

The **cumulative distribution function (CDF)** of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

For a discrete RV, the CDF is a non-decreasing step function:

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i).$$

(The notation $\sum_{x_i \leq x}$ means: add $p_X(x_i)$ for every value x_i that is $\leq x$.)

Both PMF and CDF fully describe the distribution of X — you can recover one from the other.

6.4 Example: Tossing Two Coins

Setup. Toss a fair coin twice. Let X = number of heads observed. Then $X \in \{0, 1, 2\}$.

Step 1: list outcomes. The sample space is $S = \{HH, HT, TH, TT\}$, all equally likely.

Step 2: compute the PMF.

$$p_X(0) = \mathbb{P}(X = 0) = \mathbb{P}(\{TT\}) = 1/4,$$

$$p_X(1) = \mathbb{P}(X = 1) = \mathbb{P}(\{HT, TH\}) = 2/4 = 1/2,$$

$$p_X(2) = \mathbb{P}(X = 2) = \mathbb{P}(\{HH\}) = 1/4.$$

Sanity check: $1/4 + 1/2 + 1/4 = 1$. ✓

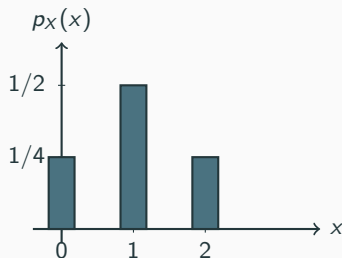
6.4 Example: Tossing Two Coins 2

Step 3: compute the CDF.

$$F_X(0) = p_X(0) = 1/4,$$

$$F_X(1) = p_X(0) + p_X(1) = 1/4 + 1/2 = 3/4,$$

$$F_X(2) = p_X(0) + p_X(1) + p_X(2) = 1.$$



PMF of X = number of heads in two coin tosses.

7. Expectation and Variance

7.1 Expected Value

Definition (discrete case)

The **expected value** (or mean) of a discrete random variable X is

$$\mathbb{E}[X] = \sum_x x p_X(x),$$

where the sum runs over all values X can take.

Notation reminder. $\mathbb{E}[\cdot]$ stands for “expectation of.” The expression $\sum_x$ means “add this up over every possible value x that X can take.” $\mathbb{E}[X]$ is also written μ_X or just μ (Greek letter “mu”).

Intuition. Run the experiment many times and average the observed values of X — you’ll get a number close to $\mathbb{E}[X]$. The expectation is the long-run average.

7.2 Linearity of Expectation

Linearity

For any constants a, b and any random variables X, Y :

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \quad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Derivation of $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$ (discrete case). Use the definition and split the sum:

$$\mathbb{E}[aX + b] = \sum_x (ax + b) p_X(x) \quad (\text{definition})$$

$$= a \sum_x x p_X(x) + b \sum_x p_X(x) \quad (\text{split + factor})$$

$$= a\mathbb{E}[X] + b \cdot 1 \quad (\text{definition; PMF sums to 1})$$

$$= a\mathbb{E}[X] + b.$$

Linearity holds **regardless** of whether X and Y are independent — one of the most useful results in probability.

7.3 Variance

Expected value tells us where X is centered. **Variance** tells us how spread out X is around its center.

Definition

The **variance** of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] .$$

The **standard deviation** of X is $\text{SD}(X) = \sqrt{\text{Var}(X)}$.

Notation reminder. $\text{Var}(\cdot)$ is “variance of” and $\text{SD}(\cdot)$ is “standard deviation of.” Variance is sometimes written σ_X^2 (sigma squared); standard deviation is σ_X .

Computational formula

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 .$$

7.4 Deriving the Computational Formula for Variance

Start from the definition and expand the square:

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\&= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] && \text{(expand the square)} \\&= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 && \text{(linearity; } \mathbb{E}[X] \text{ is a constant)} \\&= \mathbb{E}[X^2] - 2(\mathbb{E}[X])^2 + (\mathbb{E}[X])^2 \\&= \mathbb{E}[X^2] - (\mathbb{E}[X])^2.\end{aligned}$$

This formula is much easier to use in practice than the definition, since you only need to compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ separately.

7.5 Variance Properties

Useful facts

For constants a, b :

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

If X and Y are **independent**,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Note: adding a constant b shifts the distribution but doesn't change its spread, so b doesn't appear in $\text{Var}(aX + b)$. Multiplying by a scales the spread by $|a|$ in standard-deviation units (so by a^2 in variance units).

Independence is essential for $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ — unlike linearity of expectation, this rule fails without independence.

Where this shows up later. These rules drive the standard-error formulas for OLS coefficient estimates (Lecture 7).

7.6 Example: A \$1 Roulette Bet

Setup. American roulette has 38 equally likely slots: 18 red, 18 black, 2 green. Bet \$1 on red:

- Win \$1 if the ball lands red (probability $18/38$);
- Lose \$1 otherwise (probability $20/38$).

Let W = your net winnings (a random variable taking values in $\{+1, -1\}$).

Step 1: PMF of W .

$$p_W(+1) = \frac{18}{38}, \quad p_W(-1) = \frac{20}{38}.$$

7.6 Example: A \$1 Roulette Bet 2

Step 2: expected winnings.

$$\begin{aligned}\mathbb{E}[W] &= \sum_w w p_W(w) \\ &= (+1) \cdot \frac{18}{38} + (-1) \cdot \frac{20}{38} = \frac{18-20}{38} = -\frac{2}{38} \approx -\$0.053.\end{aligned}$$

7.6 Example: A \$1 Roulette Bet 3

Step 3: variance. First compute $\mathbb{E}[W^2]$:

$$\mathbb{E}[W^2] = (+1)^2 \cdot \frac{18}{38} + (-1)^2 \cdot \frac{20}{38} = \frac{18+20}{38} = 1.$$

Now apply the computational formula:

$$\begin{aligned}\text{Var}(W) &= \mathbb{E}[W^2] - (\mathbb{E}[W])^2 \\ &= 1 - \left(-\frac{2}{38}\right)^2 = 1 - \frac{4}{1444} \approx 0.997, \\ \text{SD}(W) &= \sqrt{0.997} \approx 0.999.\end{aligned}$$

Interpretation. On average you lose about 5 cents per dollar bet — the house edge. The standard deviation is about \$1, which means any individual spin's outcome is highly variable: you'll see plenty of \$1 wins and \$1 losses before the house edge becomes visible.

8. The Bernoulli Distribution

8.1 Bernoulli Distribution

Definition

A random variable X with $X \in \{0, 1\}$ follows a **Bernoulli distribution** with parameter $p \in [0, 1]$, written

$$X \sim \text{Bernoulli}(p),$$

if $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$.

Notation reminder. The symbol $\sim$ is read “is distributed as.” So “ $X \sim \text{Bernoulli}(p)$ ” means “ X follows a Bernoulli distribution with parameter p .”

Bernoulli RVs model a single **success/failure** trial: a coin toss (success = heads), a customer making a purchase (success = buys), an email being spam (success = spam).

8.2 Bernoulli Mean and Variance (Derived)

Apply the definitions directly.

Mean. The PMF is $p_X(1) = p$ and $p_X(0) = 1 - p$, so

$$\begin{aligned}\mathbb{E}[X] &= \sum_x x p_X(x) \\ &= 0 \cdot (1 - p) + 1 \cdot p = p.\end{aligned}$$

Variance. First compute $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = 0^2 \cdot (1 - p) + 1^2 \cdot p = p.$$

Then by the computational formula,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p).$$

Bernoulli summary

If $X \sim \text{Bernoulli}(p)$ then $\mathbb{E}[X] = p$ and $\text{Var}(X) = p(1 - p)$.

Where this shows up later. Bernoulli is the response distribution for binary outcomes — the basis for logistic regression (Lecture 12).

9. The Binomial Distribution

9.1 From Bernoulli to Binomial

Let $Y_1, Y_2, \dots, Y_n$ be **i.i.d.** Bernoulli(p) trials.

Notation reminder. The subscripts on $Y_1, Y_2, \dots, Y_n$ index the trial number — Y_3 is the outcome of the third trial, Y_n is the outcome of the n th. The phrase **i.i.d.** stands for “**i**ndependent and **i**dentically **d**istributed”: every trial follows the same distribution, and the trials don’t affect each other.

Define

$$X = Y_1 + Y_2 + \dots + Y_n = \sum_{i=1}^n Y_i.$$

Then X counts the number of successes (1’s) in the n trials.

Definition

X follows a **Binomial distribution** with parameters n and p :

$$X \sim \text{Binomial}(n, p).$$

9.2 Binomial PMF

Binomial PMF

If $X \sim \text{Binomial}(n, p)$, then for $k = 0, 1, \dots, n$,

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

Notation reminder. The symbol $\binom{n}{k}$ (read “ n choose k ”) counts the number of ways to choose k items from n . Its formula uses the factorial:

$$\binom{n}{k} = \frac{n!}{k! (n - k)!}, \quad n! = n \cdot (n - 1) \cdot (n - 2) \cdots 2 \cdot 1.$$

For example, $\binom{5}{2} = \frac{5!}{2!3!} = \frac{120}{2 \cdot 6} = 10$.

Where the formula comes from.

9.2 Binomial PMF 2

- $p^k (1 - p)^{n-k}$ = probability of any one specific sequence with exactly k successes and $n - k$ failures.
- $\binom{n}{k}$ = number of distinct sequences with exactly k successes (positions of those successes among n slots).

9.3 Binomial Mean and Variance (Derived)

Use linearity. Each $Y_i \sim \text{Bernoulli}(p)$, so by §8.2, $\mathbb{E}[Y_i] = p$ and $\text{Var}(Y_i) = p(1 - p)$.

Mean. Linearity of expectation:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^n Y_i\right] = \sum_{i=1}^n \mathbb{E}[Y_i] = \sum_{i=1}^n p = np.$$

Variance. Because the Y_i are **independent**, the

$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ rule applies:

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n Y_i\right) = \sum_{i=1}^n \text{Var}(Y_i) = \sum_{i=1}^n p(1 - p) = np(1 - p).$$

Binomial summary

If $X \sim \text{Binomial}(n, p)$ then $\mathbb{E}[X] = np$ and $\text{Var}(X) = np(1 - p)$.

Where this shows up later. For large n , the Binomial concentrates around np and is well-approximated by a Normal — formalized by the Central Limit Theorem (Lectures 3–4).

9.4 Example: Stocking a Ski Rental Shop

Setup. You run a ski rental shop. Of every customer who walks in, 60% want skis and 40% want a snowboard. You expect 150 customers tomorrow and you currently have 65 snowboards in stock.

Question. What is the probability you run out of snowboards?

Step 1: identify the random variable. Let X = number of customers (out of 150) who want a snowboard. Each customer independently wants a snowboard with probability 0.40 — so X is the count of successes in $n = 150$ i.i.d. Bernoulli(0.4) trials:

$$X \sim \text{Binomial}(n = 150, p = 0.40).$$

9.4 Example: Stocking a Ski Rental Shop 2

Step 2: translate the question into a probability. You run out of snowboards if more than 65 customers want one:

$$\mathbb{P}(\text{run out}) = \mathbb{P}(X > 65) = 1 - \mathbb{P}(X \leq 65).$$

Computing $\mathbb{P}(X \leq 65)$ by hand is impractical (66 terms!) — in practice we look it up using the Binomial CDF (in software). The result is:

$$\mathbb{P}(X > 65) \approx 0.179.$$

About an 18% chance of running out.

9.4 Example: Stocking a Ski Rental Shop 3

Step 3: sanity check using the mean and variance.

$$\begin{aligned}\mathbb{E}[X] &= np = 150 \times 0.40 = 60, \\ \text{Var}(X) &= np(1 - p) = 150 \times 0.4 \times 0.6 = 36, \\ \text{SD}(X) &= \sqrt{36} = 6.\end{aligned}$$

We expect 60 snowboard customers, give or take 6 (one standard deviation). Asking “what’s the probability of more than 65?” is asking how often we exceed the mean by about one SD — around 16% is a reasonable ballpark, and 17.9% lines up.

9.4 Example: Stocking a Ski Rental Shop 4

Step 4: cost-benefit comparison. If you stock 70 snowboards instead, the same calculation gives $\mathbb{P}(X > 70) \approx 0.041$ — a much safer cushion (about a 4% chance of running out). The choice between 65 and 70 boards is a classic cost-vs-risk trade-off: the extra 5 boards tie up capital, but they reduce the stockout risk from 18% to 4%.

10. Summary

10.1 What You Should Take Away

Key results from Lecture 1

- Set operations: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- Conditional probability: $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B) / \mathbb{P}(B)$.
- Independence: $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
- Law of Total Probability: $\mathbb{P}(B) = \sum_j \mathbb{P}(A_j)\mathbb{P}(B \mid A_j)$.
- Bayes' Theorem: $\mathbb{P}(A \mid B) = \mathbb{P}(B \mid A)\mathbb{P}(A) / \mathbb{P}(B)$.
- Expectation linearity: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$, always.
- Variance shortcut: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.
- Bernoulli(p): mean p , variance $p(1 - p)$.
- Binomial(n, p): mean np , variance $np(1 - p)$.

10.1 What You Should Take Away 2

Where each tool reappears in this course

- **Independence / i.i.d.** — foundation of OLS estimation (Lectures 6–9), classification (Lecture 12), resampling (Lecture 13), and trees/forests (Lecture 14).
- **Linearity of expectation** — deriving $\mathbb{E}[\hat{\beta}]$ for OLS (Lecture 7); bias–variance decomposition (Lecture 10).
- **Variance shortcut + properties** — standard errors of OLS coefficient estimates (Lecture 7).
- **Bernoulli / Binomial** — binary response models: logistic regression (Lecture 12).

Next time: continuous random variables and the normal distribution.